Quiz Questions

Q1: DIGITAL LOGIC AND MICROPROCESSOR (1)
How many minterms can be formed with two variable inputs
A. Eight minterms
B. Six minterms
C. Five minterms
D. Four minterms
Correct Answer: D. Four minterms

Q2: DIGITAL LOGIC AND MICROPROCESSOR (10)
What was the JK levels to toggle the switch?
A. J = 0, K = 0
B. J = 1, K = 0
C. J = 0, K = 1
D. J = 1, K = 1
Correct Answer: D. J = 1, K = 1

Q3: DIGITAL LOGIC AND MICROPROCESSOR (11)
How many flip-flops are required to produce a divide-by-128 device?
A. 1
B. 4
C. 6
D. 7
Correct Answer: D. 7

Q4: DIGITAL LOGIC AND MICROPROCESSOR (12)
The phenomenon of interpreting unwanted signals on J and K while Cp (clock pulse) is HIGH is called _____.
A. parity error checking
B. ones catching
C. digital discrimination
D. digital filtering
Correct Answer: B. ones catching

Q5: DIGITAL LOGIC AND MICROPROCESSOR (13)
How the master is enabled in master-slave flip-flop?
A. when the gate is LOW
B. when the gate is HIGH
C. both LOW and HIGH
D. None of the options
Correct Answer: B. when the gate is HIGH

Q6: DIGITAL LOGIC AND MICROPROCESSOR (14)
Which of the following is correct for a gated D flip-flop?
A. The output toggles if one of the inputs is held HIGH.
B. Only one of the inputs can be HIGH at a time.
C. The output complement follows the input when enabled.
D. Q output follows the input D when the enable is HIGH.
Correct Answer: D. Q output follows the input D when the enable is HIGH.

Q7: DIGITAL LOGIC AND MICROPROCESSOR (15)
In a J-K flip-flop if the clock input frequency is 20kHz what is the output at Q
A. constantly LOW
B. constantly HIGH
C. a 20 kHz square wave
D. a 10 kHz square wave
Correct Answer: D. a 10 kHz square wave

Q8: DIGITAL LOGIC AND MICROPROCESSOR (16)
If an active-HIGH S-R latch has a 0 on the S input and a 1 on the R input and then the R input goes to 0, the latch will be _____.
A. SET
B. RESET
C. Clear
D. Invalid
Correct Answer: B. RESET

Q9: DIGITAL LOGIC AND MICROPROCESSOR (17)
To made MOD-32 binary counter how many flip-flops are required?
A. 3
B. 45
C. 5
D. 6
Correct Answer: C. 5

Q10: DIGITAL LOGIC AND MICROPROCESSOR (18)
The reason behind that the delay problems faced in asynchronous counters is eliminated by Synchronous counters :
A. input clock pulses are applied only to the first and last stages
B. input clock pulses are applied only to the last stage
C. input clock pulses are not used to activate any of the counter stages
D. input clock pulses are applied simultaneously to each stage

Correct Answer: D. input clock pulses are applied simultaneously to each stage

Q11: DIGITAL LOGIC AND MICROPROCESSOR (19)
The minimum number of NAND gates required to implement the Boolean function X+XY'+XY'Z is
A. one
B. Two
C. Three
D. None of the options
Correct Answer: D. None of the options

Q12: DIGITAL LOGIC AND MICROPROCESSOR (2)
In tabulation method how the procedures is started to specify the function in terms of
A. List of minterms
B. List of maxterms
C. List of integers
D. List of terms
Correct Answer: C. List of integers

Q13: DIGITAL LOGIC AND MICROPROCESSOR (20)
The output Y of a Two-bit comparator is logic "1" whenever the two-bit input "A" is greater than the two-bit input "B". The number of combination for which the output is logic "1" is
A. 2
B. 4
C. 6
D. 8
Correct Answer: C. 6

Q14: DIGITAL LOGIC AND MICROPROCESSOR (21)
Which one of the following set of gates are best suited for 'parity' checking and 'parity' generation.
A. AND, OR, NOT gates
B. EX-NOR or EX-OR gates
C. NAND gates
D. NOR gates
Correct Answer: B. EX-NOR or EX-OR gates

Q15: DIGITAL LOGIC AND MICROPROCESSOR (22)
Which of the following code is also known as rereflected code
A. Excess-3 codes

B. Gray code
C. BCD code
D. Error code
Correct Answer: B. Gray code

Q16: DIGITAL LOGIC AND MICROPROCESSOR (23)
How '-5' is represented in hex format in 2's complement in 8-bits?
A. (FB)16
B. (7F)16
C. (FF)16
D. (FA)16
Correct Answer: A. (FB)16

Q17: DIGITAL LOGIC AND MICROPROCESSOR (24)
Which statement below best describes the function of a decoder?
A. A decoder will convert a decimal number into the proper binary equivalent
B. A decoder will convert a binary number into a specific output representing a particular character or digit.
C. Decoders are used to prevent improper operation of digital systems.
D. Decoders are special ICs that are used to make it possible for one brand of computer to talk to another.
Correct Answer: B. A decoder will convert a binary number into a specific output representing a particular character or digit.

Q18: DIGITAL LOGIC AND MICROPROCESSOR (25)
Which of the following is correct for a gated D-type flip-flop?
A. The Q output is either SET or RESET as soon as the input goes HIGH or LOW
B. The output complement follows the input when enabled
C. Only one of the inputs can be HIGH at a time
D. The output toggles if one of the inputs is held HIGH
Correct Answer: A. The Q output is either SET or RESET as soon as the input goes HIGH or LOW

Q19: DIGITAL LOGIC AND MICROPROCESSOR (3)
What labelling code is carried in the K-map
A. 8-4-2-1 binary
B. Hexadecimal
C. Gray
D. Octal
Correct Answer: C. Gray

Q20: DIGITAL LOGIC AND MICROPROCESSOR (4)
A computer has a word length of 8 bits (including sign). It is required to determine the range of integers that can be stored in the computer. If 1's complement is used to represent the numbers, then the range will be
A. -256 to 256
B. -256 to 255
C. -127 to 127
D. -128 to 127
Correct Answer: C. -127 to 127

Q21: DIGITAL LOGIC AND MICROPROCESSOR (5)
Choose the gray code form for the binary value 101011?
A. 101011
B. 110101
C. 011111
D. 111110
Correct Answer: D. 111110

Q22: DIGITAL LOGIC AND MICROPROCESSOR (6)
To design 1-of-32 decoder how many 3-to-8-line decoders are required?
A. 1
B. 2
C. 4
D. 8
Correct Answer: C. 4

Q23: DIGITAL LOGIC AND MICROPROCESSOR (7)
Consider Binary coded decimal value 0001 0010 0110 . Convert the given value to binary form.
A. 1111110
B. 1111101
C. 1111000
D. 1111111
Correct Answer: A. 1111110

Q24: DIGITAL LOGIC AND MICROPROCESSOR (8)
Consider, there are 8 doors in an automated cinema theatre. Minimum, How many number of On/Off switches are required to open any doors using digital circuit?
A. 1
B. 2
C. 3
D. 4

Correct Answer: C. 3

Q25: DIGITAL LOGIC AND MICROPROCESSOR (9)
What are the output values of the 4-bit comparator when the two input binary numbers are given as A = 1011 and B = 1010.
A. A > B = 1, A < B = 0, A < B = 1
B. A > B = 0, A < B = 1, A = B = 0
C. A > B = 1, A < B = 0, A = B = 0
D. A > B = 0, A < B = 1, A = B = 1
Correct Answer: C. A > B = 1, A < B = 0, A = B = 0